SYNTHI 100

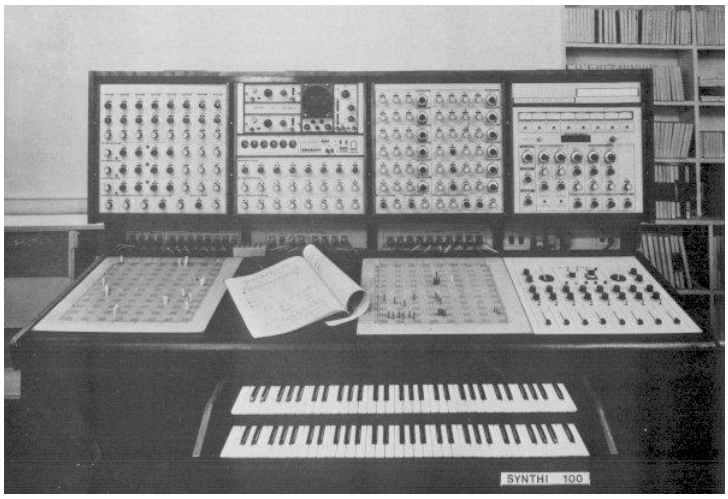
VCA and amplifier reference pages

Selected source pages for designing a dual voltage-controlled amplifier module.
Original schematic pages are included unchanged; references below use PDF page numbers.

PACKET	SOURCE PAGE / DRAWING	CONTENTS
2	Computer Synthi document, PDF p. 1	Signal-level controls and VCA context
3	Computer Synthi document, PDF p. 2	VCA noise performance and module specifications
4	Computer Synthi document, PDF p. 4	AC/DC input-amplifier specification
5	Synthi 100 service manual, PDF p. 2	Drawing-number index for circuit sheets
6	Synthi 100 schematic, PDF p. 28 — EMS 32/15, Issue E	Quad Log Modulator
7	Synthi 100 schematic, PDF p. 29 — EMS 32/16, Issue A	Hex Operational Amplifier
8	Synthi 100 schematic, PDF p. 30 — EMS 32/17, Issue F	Hex Operational Amplifiers Summing Buffer
9	Synthi 100 schematic, PDF p. 32 — EMS 32/51, Issue D	Mic. Amp

The schematic calls the VCA core a “Quad Log Modulator.” The brochure supplies operating context; the input mic amplifier is included as a separate, related amplifier circuit.

EMS Synthi 100



- This photograph shows the EMS Synthi 100. The patching matrixes are on the main desk; the right hand matrix is used for control voltages and is connected to the Digital Interface via two 60 way junctions. The knobs above the desk are used as presets when the SYNTHI is controlled by computer.

EMS Synthi 100

- The Synthi 100
- Professional Electronic Music Studio
  - The SYNTHI 100 studio has at its heart a digital sequencer, which makes conventional analogue controllers, with their gigantic array of knobs, seem old-fashioned, inaccurate and cumbersome. With a solid state storage capacity of 10,240 bits, the new sequencer is capable of precisely controlling 6 different simultaneous parameters over a sequence of 256 successive events. There are several modes of operation and full, easy to operate editing facilities, so that any or all of the 256 stored items and their time relationships may be changed without difficulty.
  - For example, two five-octave dynamically proportional keyboards are included, to operate the studio in real time, on six tracks, with the sequencer remembering what is played. This performance can then be played back backwards or forwards, at any speed, and edited to any degree of precision, prior to recording on magnetic tape.
  - The SYNTHI 100 also contains new electronic devices exclusive to EMS, such as voltage controlled slew limiters, a frequency to voltage converter and a two-output random staircase generator with controllable time and amplitude variances. Also included are a very full complement of 12 drift-free oscillators, eight dynamic filters and three transformerless i.c. ring modulators (which can be cascaded for double and triple modulation), as well as eight voltage controlled output channels with full panning facilities, eight input amplifiers, two X-Y joystick controllers, a filter bank, three elaborate envelope shapers and followers, noise generators and reverberation units. A double-beam oscilloscope and six-digit crystal-controlled counteri timer/frequency meter ensure accurate setting up and logging of parameters, and patching is by cordless pin matrices (two boards each with 60 x 60 locations).
  - In the same price range as the larger voltage controlled synthesizers, the SYNTHI 100 offers far more, and is intended for really exacting composition and realisation work in professional recording and electronic music studios, broadcasting companies and universities. First customers include the BBC, Radio Belgrade and the University of Wales.

- General Points
  - Appearance The SYNTHI 100 is presented in an afrormosia case, with padded black leather. A cupboard stand and seat are available as extras. The working surfaces are illuminated by dimmer lights, and all outputs and inputs are available behind a sliding screen in the front of the unit, while the input/output cables are channelled on to the rear. The SYNTHI 100 is designed for computer interface, and therefore all control inputs are available for direct connection to digital/analog converters on multiway connectors. The overall size base of the SYNTHI 100 is 79 in. length by 37 1/2 in. depth, the height is 33 in. and the height from the underside to the ground is 30 in.
  - Signal Levels The levels of all signal outputs available at the patchboard are controlled by panel mounted potentiometers. In fact, these potentiometers provide control voltages which operate on voltage controlled amplifiers. The advantages of this system include:
    - Constant low output impedence.
    - No fader scratch as levels are changed
    - Less conveying of signals at high impedance (where they are susceptible to crosstalk)
    - Much simplified wiring system
    - Ideal logarithmic control of level
  - In general, signal levels are about ca. 1V p-p, although most outputs can deliver much more than this. Most devices which have signal inputs are adjusted for optimum operation at this level, although it can be usually considerably exceeded without serious distortion.

- Control Voltage Levels
  - All EMS control voltage outputs are bipolar, that is to say they go both positive and negative. This allows control signals and audio signals to be interchanged, and also allows the magnitude of a control output to be adjusted without altering the mean level of the parameter controlled.

- Input Impedances
  - All input impedances are approximately 10 KOHM. This comparatively high figure was chosen as a compromise between the conflicting requirements of fan out (i.e. the number of inputs an output may drive before it runs out of power) and susceptibility to crosstalk when an input is left open circuit. Screened cable is used for all signal paths in the studio, and impedance at certain central inputs is made infinitely low so that secondary control parameters can be added without readjusting the magnitude of the primary ones.

- Noise Levels
  - The signal to noise ratio of all voltage controlled amplifiers is better than 74 dB for an input signal level which produces a T.H.D. of 3 %. This ratio is constant over the upper 30 dB of the control range. In other words, as the amplifier gain is reduced, the noise is proportionally reduced as well. When the amplifier is cut off, the noise at its output is immeasurable.

- Specification
- *Source Modules*
  - Three Voltage Controlled Audio Waveform Generators, Sine and Ramp Manual Frequency Range: Greater than 1Hz- 10kHz (extendible by voltage controls in both directions - to 0.25 Hz and 20 kHz). Sine Purity: Better than 500/0 total distortion between 10 Hz and 10KHz. Ramp Output Linearity: Departure from linearity ca. 1 % of best straight line between 10Hz and 10KHz. Voltage Control: 5v/octave. Accuracy 0.3 % departure from best straight line between 100 and 2000 KHz. Frequency Stability: Generally better than 2 % from month to month, but the oscillators usually hold their setting to within 2 cycles in a thousand during a working session. A sine shaper is included by which variable amounts of even harmonic distortion may be added.
  - Three Voltage Controlled Audio Waveform Generators, Triangle and Square. These can be varied from triangle to sawtooth ramp, and from symmetrical square to short pulse, in either polarity. Manual Frequency Range: Greater than 1 Hz to 10 KHz. Triangle Symmetry: ca.5% rise time to fall time equality. Other specifications as for sine/ramp oscillator.
  - Three Voltage Controlled Low Frequency Waveform Generators. Same details as before, but oscillators are twenty times as slow. Frequency range: Greater than 0.025 Hz (40 secs. per cycle) to 500 Hz. Voltage Control: .5v/octave. These three oscillators are intended mainly as control sources, but can be used for tone generation at the upper end of the range. All nine of the above oscillators have synchronisation inputs so that they can operate at an integral multiple of another oscillator, providing a huge variety of waveforms which can be used in additive synthesis.
  - Three Noise Generators. Variable from white (central position of colouration control) to dark or light positions (low or high pass filters). Distortion: In white position, frequency content is flat ca. 3 dB from 100 Hz- 10 KHz. We recommend several noise sources, because with different filterings more than one can be used for different purposes.
  - Dual Output Random Control Voltage Source This device produces two control voltages which move abruptly from one level to another. The distribution of levels is rectangular rather than Gaussian, and the iwo outputs are uncorrelated in level, but synchronous in time. The mean time between changes, and the variance about that mean are manually controllable. The distribution of times is rectangular, and in common with all other time controls devices in the studio, a control range of at least 1000:1 is available. Controls:
    - Amplitude Variance: Up to 2.5v symmetrically positive and negative (x2, each output separately controllable).
    - Time Range: From approximately 10 secs to 10 ms if no time variance is applied.
    - Time Variance: From equal steps to 1000:1 variance. Effect of variance control is obviously limited when time range control is at either entrance.
- *Treatment Modules*
  - Three Voltage Controlled Trapezoid Generators with Integral Envelope Shapers These devices might be described as voltage programmable four segment curve synthesizers, the output being available as two control voltages and as the modulation of an audio signal. The basic waveform produced (at each control output) has four stages:
    - 1 Delay. The output remains constant and negative for a controllable time after the generator is triggered.
    - 2 Attack. The output rises to a fixed positive value at a controlled rate.
    - 3 On. The output remains constant and positive for a controllable time after the completion of attack.
    - 4 Decay. The output falls at a controlled rate to its initial value.

    In addition, there is a second output which lags behind the first by one quarter of a complete trapezoid cycle. Thus the time set for, say, on in respect of output 1, becomes the time for attack from output 2, and so on. The amplitude and polarity of both outputs may be adjusted independently so that if they are summed (on the patchboard) any continuous four line function which ends at the value at which it starts may be produced. This arrangement gives an extraordinary flexible range of envelopes. The envelope shaper portion consists of a logarithmic voltag~ controlled amplifier permanently connected to one of the trapezoids. The overall control of each cycle of operation may be in a number of ways selected by a switch, having the following positions:
    - 1 Signal threshold. Any signal above a certain level initiates a single cycle.
    - 2 Hold sequence starts when control is positive. Decay does not start until control goes negative.
    - 3 SIngle shot positive zero crossing initiates a single cycle.
    - 4 Free run.
    - 5 Gated free run positive level allows the sequence to free run. Sequence stops at the end of a cycle when level goes negative.

    Panel Controls: Initial delay time 2 ms to 20 sec., Attack time 2 ms to 20 sec., On time 2 ms to 20 sec., Decay time 2 ms to 20 sec., (these parameters may also be voltage controlled over their entire range.) Trapezoid phase 1 output level (centre zero knob), Trapezoid phase 2 output level (centre zero knob), Signal level control, Push button (to initiate cycle), Trigger mode selector switch, Envelope shaper is logarithmic to within 3 dB over its 60 dB range., Voltage control function of time parameters is ideally exponential to within 10 % (of dependent parameter) over a range of 1000:1 Departure from ideal is gradual beyond these limits. This permits a single voltage applied to all inputs to compress the time scale.
  - Four Voltage Controlled Filter/Oscillators (Low Pass to Resonating) Operation as a sine source. Frequency Range: Greater than 5 Hz-20 KHz. Sine Purity: Better than 3 % total distortion between 10 Hz and 10 KHz. General Noise Figure for Oscillators: Spurious outputs not greater than 0.1 %. As filters, they are adjustable from Low Pass to Resonating Filters, covering the entire sonic range. Frequency Range: Greater than 5 Hz to 20 KHz. Low Pass Position: Cut off rate 12 dB for first octave and 18 dB per octave thereafter. Resonator Position: Maximum stable 'Q' factor-20. Accuracy of Exponential Voltage Control Function: ca. 1% between 100 Hz and 2000 Hz. Note: Operation as a voltage controlled oscillator is limited by the time taken to respond to an abrupt change. Maximum slew rate is about 2 ms per octave.
  - Four Voltage Controlled Filter/Oscillators (High Pass to Resonating) Similar, but complementary to Low Pass Filters.
  - One Octave Filter Bank This consists of eight resonating filters, fixed-tune one octave apart, in the range 62.5 Hz-8 KHz, seperately controllable.
  - Two Voltage Controlled Reverberation Units Each spring unit has two elements with delays of 35 and 40 ms. Maximum Reverberation Time: 2.4 seconds. Useful Frequency Range: 30 Hz-12 KHz. Voltage Control Range: ca. 2v from no reverberation to maximum reverberation.
  - Three Voltage Controlled Slew Limiters This device is a unity gain amplifier in which the output exactly follows the input at a rate whose maximum (slew) is defined by a control voltage. One application might be to interpose the device between the pitch control voltage from a keyboard and the oscillator whose pitch is to be controlled. If the key velocity voltage were then applied to the slew control input, the player could produce a glissando between any two notes that he played, the rate of glissando being controlled by his touch. Steady State Gain: 1 ca. 1 % Steady State Linearity: ca. 0.05 % (BSL) Range of Slew Control: 1 ms to 10 sec. Voltage control of slew is exponential. Note: Unlike all other devices, no output level contro; is provided as the device has unity gain.
  - Three Integral Circuit Transformerless Ring Modulators These very efficient modulators also include amplifiers, and can therefore be used in series for double or triple modulation. Maximum Input for Undistorted Output: 1.5v p-p to each input. Breakthrough with 1.5v on one input only:5mV p-p (-60 dB)
  - One 256 Event, 6 Simultaneous Parameter Digital Sequencer This machine is, in fact, a small special purpose digital computer, complete with analog to digital and digital to analog converters. It provides a sequence of control voltages which may be used on any of the devices in the studio.
    - The operation of the sequencer may most easily be described in terms of conventional music, although it must be remembered that the design by no means limits it to this kind of operation.
    - The sequencer stores 256 notes and plays each note at the correct time and for the correct duration. It simultaneously provides two voltages, one of which might be used to define pitch and the other loudness. It is capable of controlling three voices, each with duration, pitch and loudness.

- Two Five Octave Dynamically Proportional Keyboards Five octave keyboards giving precise divisions of pitch or any other controllable parameter. In the case of pitch, the range would give anything between 4 and 40 notes per octave. This is useful for microtonal work. By setting 12 notes per octave, the keyboard can be used as a normal melodic source. A second voltage output is proportional to touch -actually the velocity with which a key is struck. A third voltage switches positive when one or more keys are pressed. Note that the keyboard produces only one pitch voltage at any instant; when several notes are pressed, the voltage of the highest appears. Both the pitch voltage and dynamic voltage are remembered, even when a key is released. Keyboard voltage: 0.5V per octave maximum, accurate to better than 0.15% at all points. Dynamic voltage: ca. 1.5V depending on key velocity. Output function was synthesized to be a compromise between a strict proportionality to velocity over a range of about 100:1, and a function that would distribute seven subjectively equal increments of playing force evenly over the output voltage range. The keyboard feels most natural when the dynamics voltage is used to control a modulator over a 40 dB range.
- Eight-Way Fading/Panning Console (See Multifunction Output Amplifiers)
- 60 x 60 Pin Matrix Patchboards (7,200 Pin Locations) These patchboards allow any input to be connected to any output by the insertion of a single cordless pin. Each output is connected to a row of sixty horizontal holes. The holes appear as a square array of 3,600 (x2) cross-points, in any of which a jack may be inserted. The jacks contain resistors so that several outputs may be mixed into a single input. All device outputs are fed to the board at a low impedance, blocking any reverse signal paths. Two patchboards are provided, one intended for control signals and one for audio signals. A small number of interconnection patches between the patch-boards are hard wired, as some signals can be used in both domains. It is also possible to route external signals to the patchboards by using the jacks in the conventional way. All contacts, including the jacks have a surface coating of silver.
- Eight AC/DC Input Amplifiers Maximum distortion at rated inputs: 0.1 %. Input Sensitivities: Line Input: maximum 1.8V AC (rms) or ca. 2.5V DC. These amplifiers convert input signals level to a suitable level and impedance to feed treatment devices. The line inputs are directly coupled and are therefore suitable for both signal and low frequency or DC control inputs. Two separate microphones supplied, which can feed any two channels.
- Four External Treatment Send and Returns Provision for sending out to external echo plates and other equipment, and returning to the Studio.
- One Frequency to Voltage Converter This device accepts inputs from a variety of sources, including acoustical instruments (via a microphone or pickup and pre-amplifier) and produces a voltage proportional to the fundamental pitch of the note played. Sophisticated analog circuitry is incorporated to remove overtones, provided that their energy constitutes no more than 900/0 of the total signal. Unlike conventional frequency measuring techniques, which count the number of zero crossings of a waveform in a fixed interval of time, the converter measures the period of the signal and transforms this data to a voltage which is compatible with the other devices in the studio. The advantage of this method is that an accurate measure of the pitch can be made in a much shorter time. The output is gated into a track and hold buffer by a discriminator, which suppresses spurious outputs when the signal is dying away. A single output control adjusts the range and polarity of the output voltage.
- Two Envelope Followers These devices produce a voltage proportional to the mean level of an audio signal. The output is passed through a second order low pass filter to remove ripple while keeping a fast response. Cut off is about 50Hz. Output amplitude is adjustable by a centre zero knob to give positive or negative excursions of up to 1 volt per 6 dB.
- Items from other Manufacturers Dawe 3000 AR/6 Digital Frequency Meter Brief details as follows: Crystal frequency:100 KHz ca. 0.002 % Frequency Measurement: Range-0-1 MHz Accuracy - ca.1 digit ca. crystal accuracy Gating Time-1 ms to 10 seconds Period Measurement: Range 0-300 KHz , Time Units 1 10s to 10 ms, Gating Period - 1/7 to 1,000 cycles of input frequency Time Measurement: Range 10s - 10 secs (nearly four months) Telequipment Double Beam Oscilloscope, D43R Rack mounted double gun laboratory oscilloscope, with 6 x 8 cm display area.

- Electronic Music Studios (London) Ltd. reserve right to vary the specification and/or price of SYNTHI 100 studio without notice, should it necessary or desirable.
- source EMS London

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SYNTHI 100

DRG. No.	TITLE
32/00	Low Pass Filter Osc. Ring Mudulator Circuit
33/00	Low Pass Filter Osc. Ring Mudulator Assembly
32/01	High Pass Filter Osc. Circuit
33/01	High Pass Filter Osc. Assembly
32/02	Trapezoid Generator Circuit
33/02	Trapezoid Generator Assembly
32/04	Reverb Drive Circuit
33/04	Reverb Drive Assembly
32/05	Octave Filter Bank Circuit
33/05	Octave Filter Bank Assembly
32/06	S.R. & R.T. Oscillator & Meter Circuit
33/06	S.R. & R.T. Oscillator & Meter Assembly
32/07	Noise Generator Circuit
33/07	Noise Generator Assembly
32/08	Random Generator Circuit
33/08	Random Generator Assembly
32/09	Pitch to Voltage Circuit
33/09	Pitch to Voltage Assembly
32/10	Envelope Follower Circuit
33/10	Envelope Follower Assembly
32/11	Keyboard Controller Circuit
33/11	Keyboard Controller Assembly
32/12	Slew Limiter Circuit
33/12	Slew Limiter Assembly
32/13	O/P Channel Filter Circuit
33/13	O/P Channel Filter Assembly
32/14	Pan Summing Amps Circuit
33/14	Pan Summing Amps Assembly
32/15	Log Modulators Circuit
33/15	Log Modulars Circuit
32/16	Hex Operational Amplifier Circuit
33/16	Hex Operational Amplifier Assembly
32/17	Hex Operational Amplifier Summing Buffer Circuit
33/17	Hex Operational Amplifier Summing Buffer Assembly
32/18	Dynamic Keyboard Electronics Circuit
33/18	Dynamic Keyboard Electronics Assembly
32/19	4 Decade up/down Counter Circuit
33/19	4 Decade up/down Counter Assembly

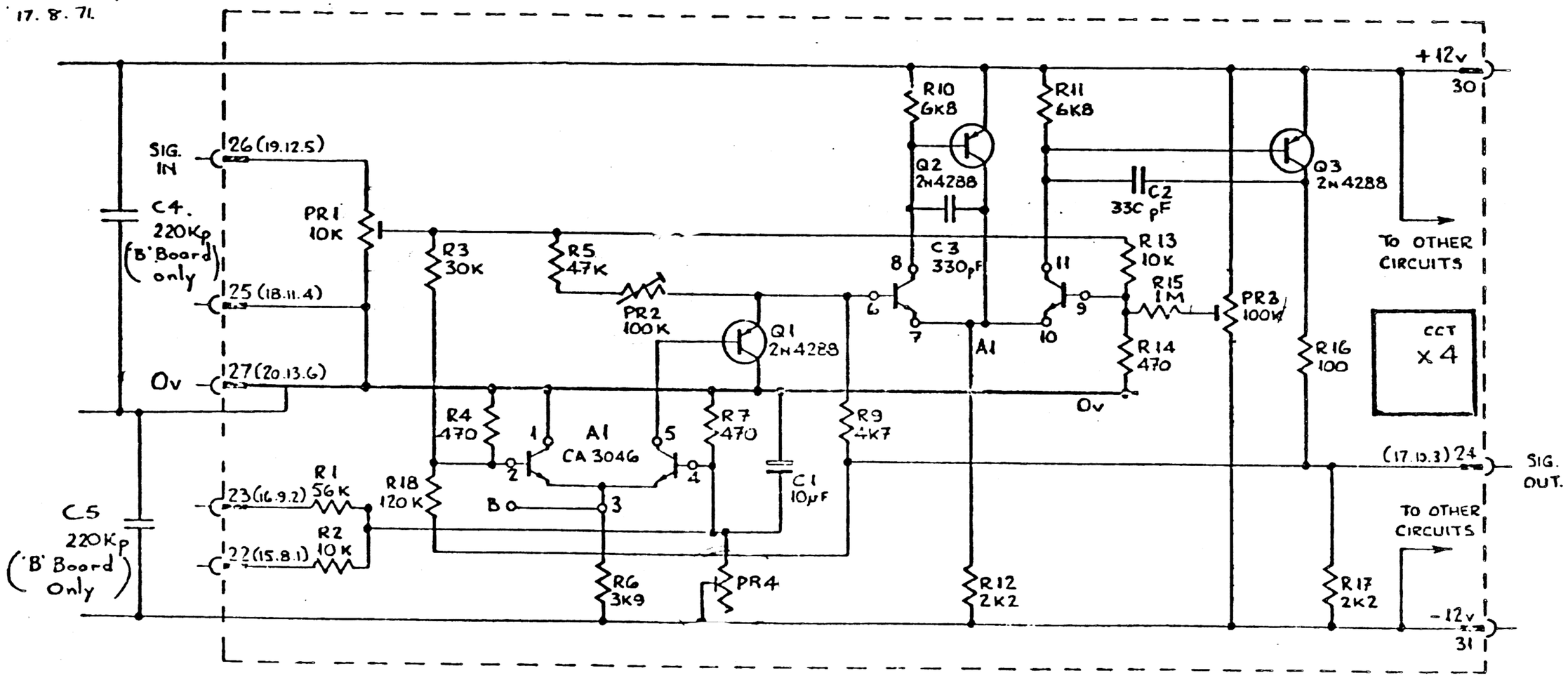
Scan by Manual Manor
http://www.markglinsky.com/ManualManor.html

B C2 WAS 1KP
C3, C4, C5 ADDED
MH 16.4.71

C R15 WAS 470K
MH 25.6.71.

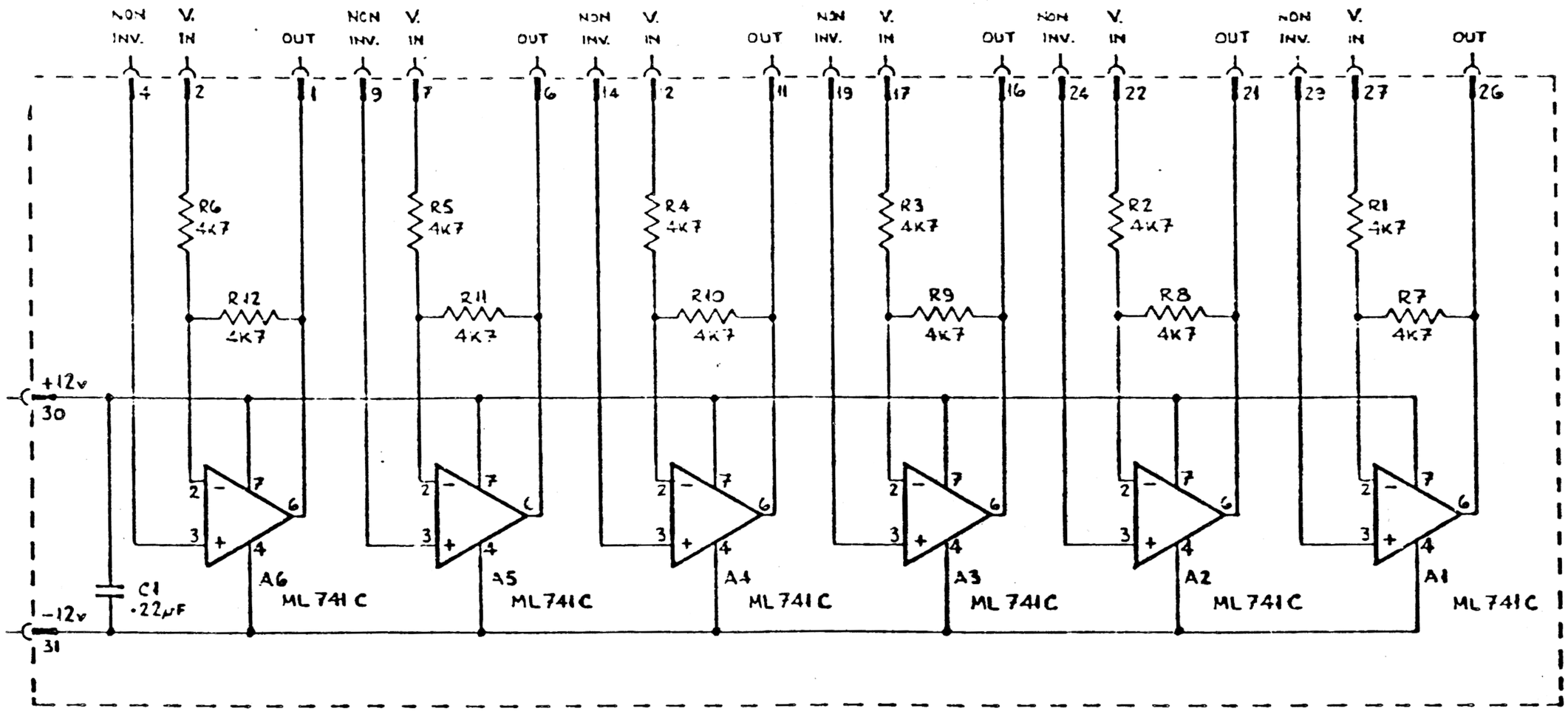
D RB Details changed.
note. filed becomes PR4.
MH 28.6.71.

E C2 & C3 d.d read 330KP
MH 17.8.71



CIRCUIT DIAGRAM QUAD LOG MODULATOR. DRG. Nº EMS32/15
ISSUE E

ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH) LTD.



CIRCUIT DIAGRAM HEX OPERATIONAL AMPLIFIER DRG. N° EMS 32/16
ISSUE A

ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH) LTD.

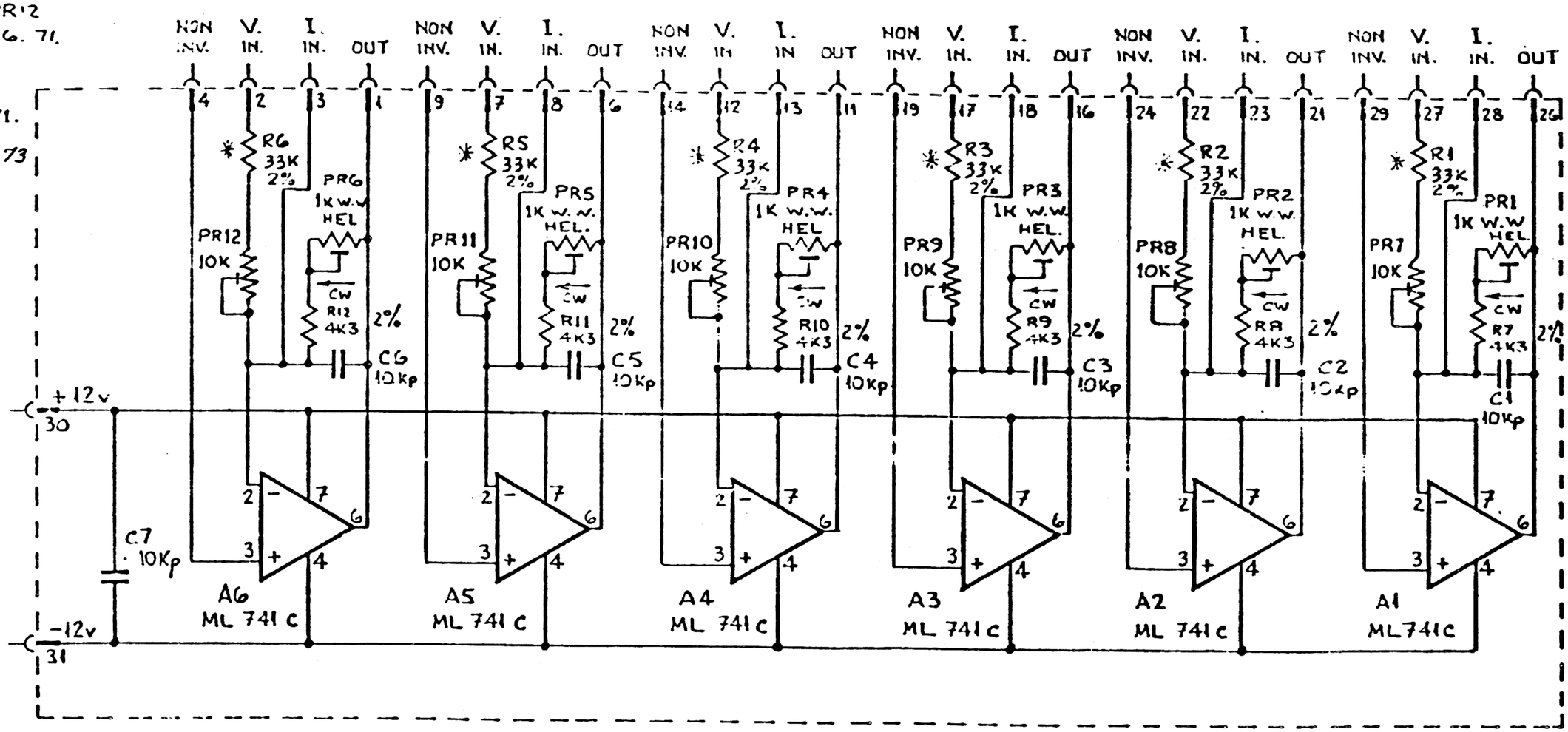
B C7 WAS .22μF
NOTES ADDED
R13 → R18 ADDED
MH 26.4.71.

C PR7 - PR12 ADDED
MH 19.5.71.

D R13 - R18 DELETED.
IIN WAS SHOWN.
COMING FROM JUNC.
OF R6 & PR12
MH 17.6.71.

E R7 to R12
WAS 3K9.
A/PC.N. 2.9.71.

F CN 49 4.4.73
DE



10K Trimpot in series with R* Osc 1→9

Replace 22K with 33K on circs 1-4 L.P. Filters

CIRCUIT DIAGRAM HEX ORERATIONAL AMPLIFIERS SUMMING BUFFER. DRGN° EMS 32/17 ISSUE F

ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH) LTD.

B R2 & R9 were 100K 27.8.71 Also C4 & C5 added.

C R8 & R9 were 270K also PR2 was 1K fixed. 4.12.72.

+12V RED

PR1 1K

C.M.R.

A1 - PIN 7

R9 270K

R3 10K

R4 10K

C2 2·5µF

C5 15µF

SIG OUT

C3 2·5µF

A1

72741P

Q1

2N5457

Q2

+

R8 270K

R1 1M

OV

OV

R2 1M.

−

C1 15µF

GAIN

PR2 1K

R5 68K

R6 68K

A1 - PIN 4

C4 15µF

−12V BLK

CIRCUIT DIAGRAM: MIC: AMP DRG: Nº EMS 32-51

ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH) LTD